TT-P-57b
January 23, 1963

SUPERSEDING
Int. Fed. Spec. TT-P-0057a(GSA-FSS)
January 20, 1962 and
Fed. Spec. TT-P-57
July 27, 1950

FEDERAL SPECIFICATION

PAINT, ZINC YELLOW-IRON OXIDE-BASE, READY-MIXED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers paint suitable for use as a priming coat on iron and steel surfaces (see 6.1).

1.2 **Classification.**

1.2.1 *Types.* The paint shall be of the following types as specified (see 6.1):

Type I. Zinc yellow-iron oxide-alkyd varnish-raw linseed oil paint (50/50 raw linseed oil-alkyd resin).

Type II. Zinc yellow-iron oxide-alkyd varnish paint.

Type III. Zinc yellow-iron oxide-phenolic varnish paint.

2. APPLICATION SPECIFICATIONS AND STANDARDS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366—Sieves; Standard for Testing Purposes.
TT-L-215—Linseed Oil, Raw (For Use in Organic Coatings).
TT-P-143—Paint, Varnish, Lacquer, and Related Materials; General Specification for Packaging, Packing and Marking.
TT-R-266—Resin, Alkyd; Solutions.
TT-R-271—Resin, Phenol-Formaldehyde, Para-Phenol
TT-T-291—Thinner Paint, Volatile Mineral Spirits (Petroleum Spirits).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Test Method Std. No. 141—Paint Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required for bidding purposes by activities outside the Federal Government are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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Military Specifications:

MIL-S-3136—Standard Test Fluids Hydrocarbon and Iso-Octane.
MIL-S-15191—Silica, Diatomaceous.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Detail requirements.

3.1.1 *Color.* The color shall be yellow for types I and II (characteristic of a mixture of lemon yellow iron oxide and zinc yellow), and red for type III (characteristic of a mixture of red iron oxide and zinc yellow).

3.1.2 *Pigment.* The extracted pigment on analysis shall conform to the quantitative requirements of table I. (See 6.2)

3.1.3 *Vehicle.* The component parts of the vehicle used in each type of paint shall be blended as follows:

3.1.3.1 *Type I.* The vehicle shall consist of an equal amount by weight of long oil alkyd resin (nonvolatile solids) and raw linseed oil together with the necessary thinners and driers. The alkyd-resin solution shall be of linseed-modified type conforming to TT-R-266, type II. Raw linseed oil shall conform to TT-L-215 (see 6.3).

3.1.3.2 *Type II.* The vehicle shall consist of a medium oil linseed soya modified meeting TT-R-266, type III, together with the necessary thinners and driers (see 6.3).

3.1.3.3 *Type III.* The vehicle shall be a straight phenolic spar varnish consisting of 15 gallons of tung oil, 10 gallons of linseed oil, and 100 pounds of a 100-percent phenolic resin plus the necessary driers and volatile thinners (see 6.4).

3.1.3.3.1 The phenolic resin for use in type III vehicle shall meet the requirements specified in TT-R-271.

3.1.3.4 *Types I, II, and III.* The vehicle shall be free from rosin and rosin deriva-

TABLE I.—*Quantitative requirements of pigment*

Ingredients	Requirements (percent by weight)					
	Type I		Type II		Type III	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Zinc yellow (CrO_3 x 2.4) ...	39	—	24	—	24	—
Iron oxide (Fe_2O_3)	18	—	37	—	31	—
Siliceous matter	—	29	—	25	—	31
Zinc oxide	14	—	14	—	14	—
Sum of the percentages of zinc yellow (CrO_3 x 2.4), zinc oxide, iron oxide, and siliceous extenders ...	90	—	90	—	90	—

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TABLE II.—*Quantitative requirements of paint*

Characteristics	Requirements					
	Type I		Type II		Type III	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Pigment, percent by weight of paint	53	61	40	45	44	48
Nonvolatile vehicle, percent by weight of vehicle	55	—	40	—	40	—
Phthalic anhydride, percent by weight of nonvolatile vehicle	10	—	30	—	—	—
Oil acids, percent by weight of non-volatile vehicle	77	—	45	—	—	—
Kauri reduction on super-centrifuged vehicle, percent	150	180	—	—	80	100
Uncombined water, percent by weight of paint	—	1.0	—	1.0	—	1.0
Coarse particles and skins (percent total residue retained on No. 325 sieve[1] based on pigment)	—	1	—	1	—	1
Consistency: Krebs-Stormer, shearing rate; 200 r.p.m: Grams	150	250	125	200	150	200
Equivalent K.U.	72	89	67	82	72	82
Weight per gallon, pounds	12.4	—	10.5	—	11	—
Drying time: Set-to-touch, hours	1	4	1/2	2	1/2	2
Dry-through, hours	—	24	—	16	—	16
60° specular gloss	—	60	—	40	—	50
Fineness of grind	5	—	5	—	5	—
Flash point (Pensky-Martens) °F.	86	—	86	—	86	—

[1] Conforming to Federal Specification RR-S-366.

tives. The test for rosin under 4.3.1 shall be negative. The vehicle may contain additional agents such as anti-oxidants and wetting aids.

3.2 Quantitative requirements.

3.2.1 *Paint.* The paint shall meet the quantitative requirements specified in the applicable portions of table II.

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3.3 **Qualitative requirements.**

3.3.1 *Condition in container.* The paint shall not show excessive settling in a freshly opened full can, and shall easily be redispersed with a paddle to a smooth, homogeneous state. The paint shall show no curdling, livering, caking, or color separation, and shall be free from lumps and skins.

3.3.2 *Dilution stability.* The paint shall remain stable and uniform for 48 hours after reduction of five parts by volume of the package material with one part by volume of mineral spirits conforming to TT-T-291.

3.3.3 *Brushing properties.* The paint as received shall work easily, and flow out readily under the brush with no tendency to pull. It shall cover evenly on a vertical, smooth metal surface without running, streaking, or sagging when applied at a spreading rate of 500 square feet per gallon.

3.3.4 *Spraying properties.* The paint when reduced with not more than one part by volume of mineral spirits conforming to TT-T-291 to five parts of the package material, shall spray satisfactorily, show no tendency to orange-peel, sag, creep, or run, and shall show satisfactory spraying properties in all other respects.

3.3.5 *Appearance.* The paint shall dry to a smooth, uniform finish free from roughness, grit, unevenness, or other surface imperfections.

3.3.6 *Flexibility and adhesion (type II).* The paint shall show no cracking when tested in accordance with method 6221 as modified in footnote 3 of 4.3.1. Likewise the paint shall adhere tenaciously to the metal panel when tested in accordance with method 6304 in 4.3.1. It shall be difficult to lift any film from the surface, and the cut shall show a fine feathered edge down to the metal.

3.3.7 *Immersion resistance.*

3.3.7.1 *Cold water (types II and III paint).* The paint film shall show no wrinkling and not more than slight blistering immediately after removal from the water. The film shall show not more than very slight whitening, and not more than slight dulling or softening when examined 2 hours after removal. (See 4.3.2.1.)

3.3.7.2 *Hydrocarbon resistance (type II paint).* A film of paint prepared and tested as in 4.3.1 and 4.3.2.3 shall show no blistering or wrinkling immediately after removal of the panel. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, adhesion, and general appearance from a panel similarly prepared at the same time but not immersed.

3.3.8 *Skinning.* The paint shall not skin within 48 hours in a three-quarter-filled closed container.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 **Sampling and inspection.** Sampling and inspection shall be in accordance with methods 1011, 1021, and 1031 of Fed. Test Method Std. No. 141 as applicable.

4.3 **Tests.** Tests shall be made in accordance with the applicable methods as specified hereinafter.

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4.3.1 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141:

Test	Method No.
Condition in container	3011
Weight per gallon	4184
Percentage of pigment	4021
Isolation of vehicle	4032
Nonvolatile in vehicle	[1] 4041
Drying time	4061
Uncombined water	4081
Coarse particles and skins	4092
Skinning	4141
Kauri reduction	[2] 4152
Consistency	4281
Flash point	4293
Brushing properties	4321
Spraying properties	4331
Dilution stability	4203
Fineness of grind	4411
Rosin and rosin derivatives	[2] 5031
Immersion resistance	6011 and par. 4.3 herein
Specular gloss	6101
Knife test	6304
Flexibility	[3] 6221
Phthalic anhydride	7021
Oil acids	7031
Analysis of extracted pigment	7331

[1] The clear vehicle shall be collected in accordance with method 4032. Use a 1.5 gram sample for types I and III and heat for 3 hours. For type II use 0.8 to 1.2 grams and heat for 1 and 2 hours as specified in method 4041, note 1.

[2] Make the test on a portion of the isolated vehicle.

[3] Apply type II paint to a flat tin panel (method 2012) with a 0.002 inch (approximately 0.004 inch gap-clearance). Bird film-applicator or any other doctor blade which produces a film of the same thickness as that produced by the Bird blade and allow to air dry for 18 hours at room temperature. Bake for 5 hours at 105° ± 2°C. (221° ± 4°F.). Condition the panel for 15 minutes at 23° ± 1.1°C. (73.5° ± 2°F.) and then bend at that temperature over a 1/4-inch mandrel. Examine for cracks in strong light at a 7-diameter magnification. Apply knife test (method 6304).

4.3.2 *Immersion resistance tests.*

4.3.2.1 *Cold-water-resistance test (type II paint).* Draw down a 2-inch film, using a 0.002 inch (approximately 0.004 inch gap-clearance) Bird film applicator, or any other doctor blade which produces a film of the same thickness as that produced by the Bird blade on a 3- by 5-inch tin panel (method 2012) and air-dry for 48 hours. Immerse in distilled water at 23°±1.1°C. to a depth of 3 inches for 18 hours. Remove the panel and examine visually for wrinkling and blistering. Air-dry the panel for 2 hours and examine visually for whitening and dulling. Softening shall be determined by scratching the unimmersed and the immersed portion of the test panel with a knife blade. (See 3.3.7.1.)

4.3.2.2 *Cold-water-resistance test (type III paint).* Apply two coats of type III paint to two solvent-cleaned steel panels prepared in accordance with method 2011 of Fed. Test Method Std. No. 141 that have previously been backed with two coats of the same paint, by brushing (method 2141) allowing 24 hours air-drying between coats. After the second coat has air-dried for 96 hours, dip the edges of the panels (about 1/4 inch) in a water-resisting varnish and air-dry for an additional 24 hours. Place the panels in a beaker containing approximately 3 inches of distilled water at room temperature (21° to 32°C.; 70° to 90°F.) and allow to remain for 14 days. Remove, allow to dry for 2 hours at room temperature, and examine the coating for defects described in 3.3.7.1.

4.3.2.3 *Hydrocarbon resistance (type II).* Prepare a film of the paint as in 4.3.2.2 and

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air-dry 96 hours. Do not wax or coat the exposed metal surfaces. Immerse the panel for 4 hours in a hydrocarbon fluid conforming to MIL-S-3136, type III, in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period remove and examine for compliance with 3.3.7.2.

4.3.3 *Appearance* (*all three types*). Examine the panels prepared for brushing and spraying properties (see 4.3.2.1).

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging, packing, and marking.** The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.8). The paint shall be furnished in 1-gallon metal cans or 5-gallon steel pails as specified (see 6.8).

6. NOTES

6.1 **Intended use.**

6.1.1 Type I primer paint is intended for use on bridges and similar rough structural steel, either as a shop primer or a field primer, when good durability to atmospheric exposure is required. It is characterized by easy brushing on rough steel, good penetration into voids (rather than bridging the depressions), ability to cling to sharp edges (such as bolt heads), and has good weather resistance. The vehicle for this paint consists of a blend of 50 percent long oil alkyd resin and 50 percent raw linseed oil (not bodied) with thinners and driers. The presence of the free linseed oil is desirable for the purpose of obtaining intimate bonding with the metal surface due to the wetting characteristics of the oil.

6.1.2 Type II primer paint is not intended for use on rough steel such as structural steel (bridges, tanks, etc.). It is intended for use on smooth and thoroughly cleaned steel, particularly as an industrial primer for use on smooth, steel articles, fabricated at the factory, such as refrigerators, railroad cars, automobiles, etc. It may also be used as a general maintenance primer for touch-up work on smooth, thoroughly cleaned steel where rapid drying is essential. Primarily, it is intended for air-drying, but may be baked (45 minutes at 250° F.). The vehicle in this primer is a straight oil-modified (linseed soya oil) alkyd varnish (medium oil length, TT-R-266). It is essential that paints of this type be applied to exceptionally clean surfaces, since due to their fast-drying properties and reduced oil content, they do not have the surface wetting properties which characterize paints containing free linseed oil. Thus, because structural steel (tanks, bridges, etc.) is rarely cleaned free of corrosion products, this paint is not generally intended for such use, although with due consideration to surface preparation it can be so used for spot-priming.

6.1.3 Type III paint is intended for use as a primer on iron and steel surfaces, particularly those subjected to severe humid conditions or fresh-water immersion. Water storage tanks (not potable), structural work on dams, structural metal and equipment in bottling plants, laundries, and mines are typical of the steel surfaces for which this paint is particularly effective as a protective coating. The same precautionary steps should be taken to insure a clean surface for application as required for type II paint. The use of phosphoric acid pretreatment solutions similar to those contemplated by TT-C-490 is a valuable adjunct in painting clean structural steel surfaces subject to immersion in water. The pretreatment solution is particularly valuable when used in conjunction with type III paint. The phosphoric acid solution should be applied to surfaces which have been thoroughly cleaned to base metal by sand-blasting.

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6.2 As a guide to the producers, the following pigment formulations are suggested in making up the three types of paint covered by this specification.

Pigment composition

Pigment	Requirements (percent by weight)						Ingredient specification
	Type I		Type II		Type III		
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
Zinc yellow ..	40	—	25	—	25	—	TT-P-456
Yellow iron oxide	22	—	45	—	—	—	TT-P-458
Red iron oxide	—	—	—	—	41	—	TT-P-405
Zinc oxide ...	15	—	15	—	15	—	TT-P-463
Magnesium silicate	—	23	—	15	—	13	TT-P-403
Diatomaceous silica	—	—	—	—	5	10	MIL-S-15191

6.3 Typical vehicle composition for varnishes meeting types I and II are as follows:

Vehicle composition

Ingredients	Requirements (percent by weight)			
	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Raw linseed oil TT-L-215	29	—	—	—
Alkyd resin solids	29	—	40	—
Volatile thinner (TT-T-291) and drier (TT-D-643)	—	42	—	60

6.4 A typical formula for varnish meeting type III is as follows:

	Pounds	*Gallons*
Phenol formaldehyde resin (TT-R-271)	18.7	1.87
Raw linseed oil (TT-L-215) ...	14.4	1.86
Mineral spirits (TT-T-291) ...	35.7	5.44
High flash naphtha	9.3	1.28
Tung oil (TT-T-775)	21.9	2.80
Total	100.0	13.25

Heat one-half of the resin and the raw linseed oil to 560°F. in 30 minutes. Hold at 560°F. for 1 hour and check with the remaining resin. Stir until melted (430°F. for 5 minutes), add tung oil (300°F.). Reheat to 465°F. in 20 minutes and hold to 10 to 25 minutes for body. Remove from fire and thin to a viscosity of E (Gardner-Holdt).

6.5 The composition of the three types of paint is so chosen that the pigment volume (PV) lies between 35 and 40 percent.

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6.6 The color of types I and II paints is yellowish; that of type III, reddish.

6.7 The three types of paint covered herein have good hiding power (because of the presence of iron oxide). Thus there may be a tendency to apply films that are too thin for good durability. The minimum thickness of the dry paint-film should be approximately 1.5 mils per coat.

6.8 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of material.
(*c*) Level of packaging, packing and marking required (see 5.1).
(*d*) Whether 1-gallon or 5-gallon containers are required (see 5.1).

6.9 Paint covered by this specification should be purchased by volume (231 cubic inches at 15.5°C. (60°F.) equals 1 U. S. gallon).

6.10 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Paints, stains or varnishes, not otherwise indexed by name, liquid.
Carload minimum weight 36,000 pounds.

Motor:

Paints, stains or varnishes, not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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